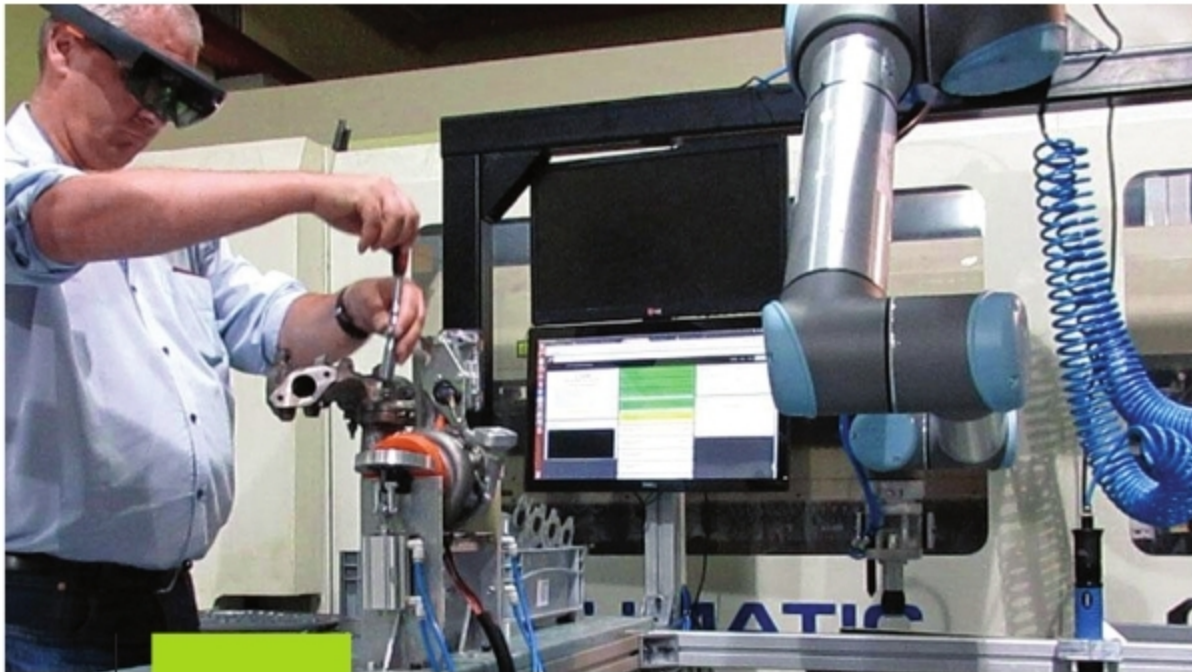




How AR and VR are Leading to a Better Future



AR and VR platforms for industrial processes (Credit: www.eonreality.com)

One of the biggest benefits in an augmented reality (AR)- and virtual reality (VR)-enabled environment is the ability to innovate products and services in a completely new way.

Adoption of augmented reality (AR) and virtual reality (VR) technologies in education, healthcare and industries has led a new foundation for digital transformation. Students and doctors can learn in new ways by experiencing 3D views of their designs. Employees can interact with machines with the help of AR- and VR-guided training materials. The entertainment industry is also exploring AR and VR content for an exciting cinematic experience. These experiences will further improve with advancements in AR- and VR-based technologies and assistive tools.

Many big manufacturing and automotive industries are deploying AR and VR platforms to achieve higher efficiency and error-free training. For

example, Blue Star has applied an AR/VR platform for inspection and manufacturing operations. And, Royal Enfield has applied it for marketing purposes. Similarly, Volvo's manufacturing plants have deployed such a platform for engine quality control and assurance.

Earlier, training new operators on complex inspection procedures took five weeks, which added to the overall cost. Now, using Vuforia AR experience, operators can quickly recall the most up-to-date configurations in 3D to ease the burden of sorting through stacks of paper, creating gains in productivity, quality control and overall process efficiency.

In the Industry 4.0 world, youth with skills and competencies in AR and VR will have a unique advantage in the job market. AR and VR comprehensive

sessions will enable enterprises discover how these technologies can change the process, and reduce cost of training and product design.

VR, AR and mixed reality (MR) can replicate the real environment using advanced processing, computing, sensing and display technology. One of the biggest benefits in a VR- and AR-enabled environment is the ability to innovate products and services in a completely new way.

Role of AI and ML in AR-VR integration

Artificial intelligence (AI) and machine learning (ML) technologies help reduce human error rates through AR and VR simulations of dangerous tasks and environments. India's skill development, training and learning space will create thousands of job opportunities in gaming, immersive advertising, creative media, movies, and AR and VR. Centres of excellence (CoEs) in this area will create thousands of job opportunities in the gaming and creative media spaces.

ML enables the machine to learn from the various tasks it performs to improve itself. AI ensures that the machine applies that learning algorithm best based on different situations. For example, a system could be trained using deep learning to recognise more complex scenarios or components. A camera's view of parts in an engine could suggest a repair to the technician, with instructions pinpointed on the image.

Also, an AR display that uses AI image recognition can streamline complex maintenance in aviation.

AI-based personalised shopping is another example that utilises VR devices.

AR and ML can be used together to solve business challenges and